

Search for Li-electrochemical activity and Li-ion conductivity among lithium bismuth oxides

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ABSTRACT

Previously reported lithium bismuth oxides, LiBiO_2 , LiBiO_3 , Li_3BiO_3 , Li_3BiO_4 , Li_5BiO_5 , and Li_7BiO_6 were prepared either via ceramic or hydrothermal synthesis methods, and their feasibility as lithium battery cathode materials was tested. None of the oxides showed any desirable electrochemical activity at higher potentials, except for the limited capacity observed during the first discharge in LiBiO_3 . In contrast, these phases uptake large amounts of Li^+ via conversion reactions when cycled down to zero volt, but the reversibility is poor. The transformation of LiBiO_3 to LiBiO_2 and Li_3BiO_3 to Li_3BiO_4 observed via X-ray diffraction showed absence of intermediate phases with mixed $\text{Bi}^{3+}/\text{Bi}^{5+}$ oxidation states. Among the materials studied, the highest ionic conductivity of $3.8 \times 10^{-8} \text{ S/cm}$ at 25°C was measured for LiBiO_2 . The Li^+ conduction pathways deduced from bond valence energy landscape approach suggest the best Li diffusion characteristics in Li_7BiO_6 among the Li-Bi-O phases studied.

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Efficient energy production, storage, and distribution are major global challenges, and Li-ion batteries with their wide range of applications are a promising solution [1]. However, the ability of Li-ion batteries to successfully address the growing energy storage demands relies on the development of new and efficient electrode materials. Such new materials must have high power and energy densities, while simultaneously being inexpensive and sustainable. Further, candidates for Li-ion electrodes must have sufficient electronic and ionic conductivities as well as an open structure for the insertion-extraction of Li-ions [2].

Currently available Li-ion batteries use transition metal oxide-based cathodes; either layered (LiMO_2) or spinel (LiM_2O_4) with $\text{M} = \text{Co}, \text{Ni}, \text{Mn}$. While these materials have the advantage of high redox voltages, they are plagued by high cost (Ni and Co) and limited chemical stability. Hence, there is a need for stable yet inexpensive alternatives. In fact, the search for such materials over the years has led to the discovery of a large number of Fe-based polyanionic compounds, including LiFePO_4 , which is now commercially used. Although these polyanionic materials are desirable for their low costs, their low capacities due to high

molecular weight makes them less ideal. Therefore, there still exists a need for new and affordable high energy density electrode materials.

The energy density of electrode materials can be enhanced either by increasing their voltage or specific capacity. The latter of which can be achieved by designing low molecular weight insertion compounds or compounds with the ability to accept more than $1e^-$ per 3d metal. While significant efforts have been dedicated to increasing capacities by decreasing molecular weight, literature search suggests very little exploration of materials with more than $1e^-$ per 3d metal. Among those previously explored are vanadium- and manganese-based compounds, which exhibit a wide range of oxidation states ($\text{V}^{5+} \rightarrow \text{V}^{2+}$, $\text{Mn}^{4+} \rightarrow \text{Mn}^{2+}$) [3]. One such example is $\text{Li}_2\text{MnSiO}_4$ phases with $\text{Mn}^{2+}/\text{Mn}^{4+}$ redox couple that have been shown to remove 2 Li^+ , but the inability to form MnSiO_4 (in agreement with DFT calculations) has limited their pursuit [4]. In contrast, although LiVPO_4F has been successfully shown to display a two electron process, the large voltage interaction domain ($> 2 \text{ V}$) associated with successive $\text{V}^{2+}/\text{V}^{3+}$ and $\text{V}^{3+}/\text{V}^{4+}$ redox couples has made it less suitable for battery applications [5,6]. We are therefore interested in identifying new compounds within which a $2e^-$ process can occur at a single voltage as a result of the favorable electronic configuration of constituent elements.

Members of column V in the periodic table (Bi, Sb, As) with s^2 lone pair electrons open up the possibility of having oxidation state changes

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in steps of two. In fact, Bi-based oxides are extensively studied for high T_c superconductors [7,8], and several compounds (e.g. BaBiO_3) with mixed $\text{Bi}^{3+}/\text{Bi}^{5+}$ oxidation states have been reported [9,10]. These mixed-valence compounds could present disproportionation reactions and consequently prove to be good candidates for understanding $2e^-$ redox processes. However, little is known regarding their electrochemical behavior, especially with respect to Li-ion batteries.

Recent work by Sathiyamoorthi et al. [11] on layered LiBiO_2 showed complete removal of Li from the structure and a capacity of 110 mAh/g. While the authors do not discuss the oxidation mechanism responsible for such high capacities in the Bi-based material, one of the possibilities is a $2e^-$ transfer resulting from a disproportionation reaction ($2\text{LiBiO}_2 \rightarrow 2\text{Li} + \text{Bi}^{3+}\text{Bi}^{5+}\text{O}_4$). Such an observation motivates the study of other Li-Bi-oxides, which have Bi in +3 oxidation state. Among them, oxides that can reversibly exchange 2Li^+ during the $2e^-$ redox process would be of particular interest. Accordingly, Li_3BiO_3 with the removal of 2Li^+ to promote the oxidation of Bi^{3+} to Bi^{5+} would have a theoretical capacity of 192.9 mAh/g, which is comparable to that of commercially available LiFePO_4 (170 mAh/g). Hence, we studied all the reported Li-Bi-oxide phases (LiBiO_2 [12,13], LiBiO_3 [14], Li_3BiO_3 [15], Li_3BiO_4 [16], Li_5BiO_5 [17,18], and Li_7BiO_6 [19]) for Li-ion battery cathode applications. Here we discuss their synthesis, electrochemical activity, and Li-ion conductivity. We also include a short discussion on some of the corresponding Li-Sb oxide phases to provide an insight into their feasibility as cathode materials.

1. Synthesis

A 5 g intimate mixture of $\text{Bi}_2\text{O}_3:\text{Li}_2\text{O}$ in a molar ratio corresponding to the Bi:Li ratio of the desired final oxide phase with 10 mol% excess of Li_2O was prepared by ball milling for 30 min on a Spex Miller®. The stainless steel milling jars were filled and emptied in an Ar-filled glove box. The milled mixture was pressed into a pellet (diameter 13 mm $\times$ 3 mm thickness) and heated in a quartz crucible in air for 24 h at 525 °C to make LiBiO_2 and Li_3BiO_4 , and at 650 °C to make Li_5BiO_5 and Li_7BiO_6 samples. Li_3BiO_3 was prepared by heating under Ar flow for 16 h at 450 °C.

The LiBiO_3 samples were prepared via hydrothermal methods using a slight variation of the procedure described by Kumada et al. [14]. A 0.5 g mixture of 1:12 molar ratio of $\text{NaBiO}_3 \cdot n\text{H}_2\text{O}$ to LiOH was added to 15 mL of distilled water and heated to 120 °C for 48 h in a Teflon-lined Parr bomb. The resulting mixture was washed with water and dried at 80 °C in a vacuum oven for 2 h.

Similar to the Li-Bi-O materials, several of the reported Li-Sb-O phases (LiSbO_2 , Li_3SbO_4 , LiSbO_3 , Li_5SbO_5) were prepared by ceramic process starting from Li_2O and Sb_2O_3 precursors and heating in air at a rate of 100 °C/h to between 450 °C and 600 °C overnight.

2. Characterization

2.1. X-ray powder diffraction

X-ray powder diffraction measurements (XRD) were conducted using Bruker D8 diffractometers with either a $\text{Co-K}\alpha$ ($\lambda_1 = 1.78897 \text{ \AA}$, $\lambda_2 = 1.79285 \text{ \AA}$) or $\text{Cu-K}\alpha$ ($\lambda_1 = 1.54056 \text{ \AA}$, $\lambda_2 = 1.54439 \text{ \AA}$) radiation source. The phase purity of the samples was confirmed by a Le Bail fit of the resulting patterns conducted using the FullProf program [20].

2.2. Thermogravimetric analysis

TGA analysis was conducted using a Netzsch STA 449C apparatus (TGA-DSC) and heating samples in alumina crucibles either under Ar or O_2 flow at 10 °C/min from room temperature to 600 °C.

2.3. Electrochemical testing

The working electrode was prepared by mixing the active electrode material with 25 wt% SP carbon and ball milling for 20 min on a Spex® miller. The material was then tested in Swagelok®-type half cells vs. Li^+/Li^0 using Whatman GF/D borosilicate glass fiber separator and LP30 electrolyte. The cells were assembled in an Ar-filled glove box and generally contained 8–10 mg of active material per cell. Galvanostatic charge–discharge tests were conducted at room temperature (20 °C) using a “Mac-Pile” or a VMP system (Biologic S.A., Claix, France).

2.4. Ionic conductivity measurements

Impedance spectroscopy was used to evaluate ionic conductivity of the Li-Bi-O phases. Samples were pressed into pellets (13 mm diameter $\times$ 1.5 mm thickness) and sintered for 10 h at 450 °C either in air or under Ar (Li_3BiO_3), and carbon paper discs were used to make electrical contact (the contact between C and the pellet was ensured by pressure). Measurements were conducted over a range of frequencies (1 MHz to 1 Hz, using 11 points per frequency decade) as a function of temperature (every 25 °C from 250 °C to 450 °C), in an in-house conductivity setup. The linearity of the system was checked by modifying AC amplitude from 50 to 350 mV before conducting *in situ* measurements. Under the conditions of the experiment, the pellet contribution is described by a semi-circle, and the total resistance of the pellet is attributed to the diameter of the measured semi-circle. Using the measured resistance (R) and dimensions of the pellet [surface area (S) and thickness (e)], the conductivity (σ) values at different temperatures was calculated using the $\sigma = \frac{1}{R} \times \frac{e}{S}$ relationship.

3. Results and discussion

Our study was originally motivated by a report from Sathiyamoorthi et al. [11] showing good electrochemical cycling for LiBiO_2 , but their published procedure for combustion synthesis was not reproducible. In fact, none of the X-ray powder patterns of samples prepared using their synthesis procedure matched those reported for LiBiO_2 , thereby raising doubts about the structure and composition of the material used in their study. However, LiBiO_2 is a well-characterized material and was easily prepared using standard ceramic synthesis [13].

Expanding to other reported Li-Bi-oxides, Li_3BiO_3 , Li_3BiO_4 , Li_5BiO_5 , and Li_7BiO_6 phases were prepared via ceramic synthesis. Both LiBiO_2 and Li_3BiO_4 were prepared at 500 °C while Li_5BiO_5 and Li_7BiO_6 required heating to 650 °C in air. The Li_3BiO_3 phase was obtained by heating to 450 °C under Ar flow, while LiBiO_3 was made using hydrothermal methods. The quality of each sample was ascertained by a Le Bail fit to their X-ray powder diffraction patterns (Fig. 1). The resulting lattice parameters along with relevant crystal structure information are provided in Table 1, and a crystal structure representation for each phase is shown in Fig. 2.

We began our study by examining the electrochemical properties of LiBiO_2 , which has a layered structure suitable for Li insertion/extraction. However, when LiBiO_2 was cycled in Swagelok-type Li-half cells, there was no electrochemical activity observed (Fig. 3a). Even with optimizing the carbon loading and cycling conditions, we did not see any electrochemical activity and hence failed to reproduce the electrochemical cycling previously reported by Sathiyamoorthi et al. [11]. However, our results do not come as a complete surprise when we take a closer look at the structure of LiBiO_2 where Li atoms are trapped between layers of Bi-O tetrahedra. This is unlike Li atoms in Li-3d transition metal-based layered oxides, where the Li atoms sit within the van der Waals gap of the M-O layers.

Surveying the other Li-Bi-O compounds, the only other phase containing Bi^{3+} is Li_3BiO_3 , and unlike LiBiO_2 , it has multiple Li^+ per formula unit, which could enable a reversible $2e^-$ redox process. However, it

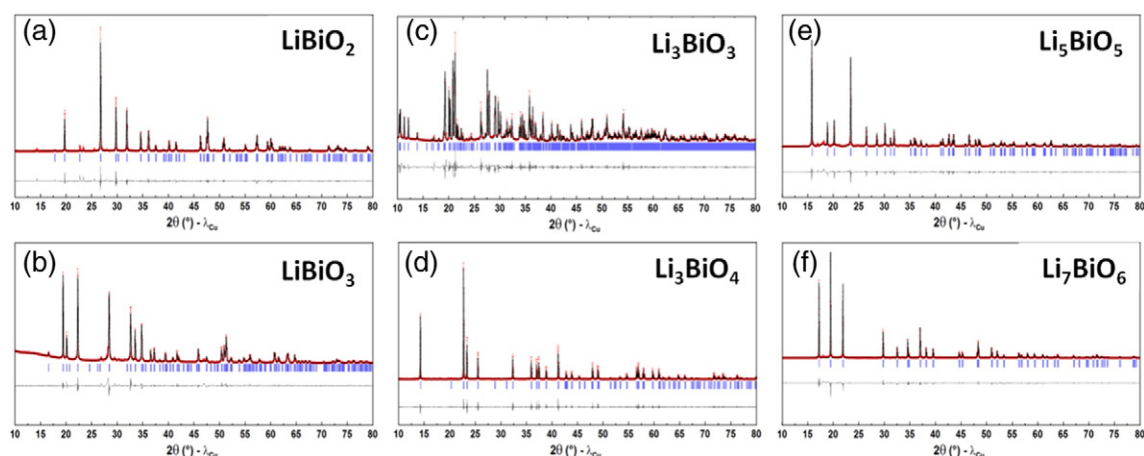


Fig. 1. Le Bail fits of X-ray powder diffraction patterns of (a) LiBiO_2 , (b) LiBiO_3 , (c) Li_3BiO_3 , (d) Li_3BiO_4 , (e) Li_5BiO_5 , and (f) Li_7BiO_6 samples prepared and examined in the study.

does not have an open crystal structure for Li-insertion/extraction, and accordingly, the material failed to show notable electrochemical activity when cycled vs. Li^+/Li^0 (Fig. 3c). The remaining Li-Bi-O phases (LiBiO_3 , Li_3BiO_4 , Li_5BiO_5 , and Li_7BiO_6), which all contain Bi^{5+} , were cycled starting on discharge vs. Li^+/Li^0 , but again none of them showed any activity upwards of 2 V vs. Li^+/Li^0 (Fig. 3 b, d, e, f). However, all the oxide phases studied (Bi^{3+} and Bi^{5+}) underwent classical conversion reactions at lower potentials (Fig. 4), and irrespective of the starting oxide phase, a Li uptake amount between 8 and 9 was observed. The conversion products were deduced to be a mixture of Li_2O and Li_xBi alloys. In fact, the specific conversion process has been previously reported by several groups to occur via the following reaction: $\text{LiBiO}_2 + 6 \text{Li}^+ \rightarrow \text{Li}_3\text{Bi} + 2\text{Li}_2\text{O}$ [21–23]. Due to the poor reversibility of the Li-Bi-O/Li cells, detailed analysis of the conversion products was not pursued.

In case of LiBiO_3 , during the first discharge, some capacity was noticed around 3.4 V prior to the onset of conversion reaction. To understand the origin of this capacity, we examined the first discharge-charge cycle using *in situ* X-ray diffraction (Fig. 5). The diffraction data collected at a very slow discharge rate of C/60 suggest a biphasic process associated with the growth of new peaks at the expense of pristine LiBiO_3 . The resulting diffraction patterns were indexed to the same orthorhombic cell as LiBiO_3 (space group $Pccn$) but with lattice parameters $a = 9.12 \text{ \AA}$, $b = 5.14 \text{ \AA}$, $c = 10.79 \text{ \AA}$, and $V = 506.8 \text{ \AA}^3$. These values show a 9% cell-volume increase compared to pristine LiBiO_3 , indicating formation of a lithiated Li_xBiO_3 phase ($x > 1$). However, the precise value of x is currently not known. Upon charging, the LiBiO_3 electrode material quickly lost crystallinity and became amorphous.

The lack of electrochemical activity in certain materials could be related to their low ionic conductivity. A recent and elegant approach to evaluate whether a compound is an intrinsically good ionic conductor

has been developed and successfully applied to energy materials and ionic conductors by S. Adams [24–26]. The method has now been implemented in the BondSTR program of the FullProf Suite [20], and relies on bond valence sum maps and bond valence energy landscapes (BVEL) obtained using soft bond valence sum parameters. BVELs are obtained from soft-BVS parameters by transforming valence into energy units, and using a Morse-type potential for the attractive part and a screened Coulomb potential for the repulsive part [25]. The crystallographic unit cell of interest is then divided along a 3-dimensional grid with adjacent points separated by 0.1 Å, and on each point, an imaginary Li^+ ion is placed (previously existing Li-species are removed from the list of ions prior to the calculations), and its bond valence sum and bond valence sum energy are calculated (here we used a distance of 8 Å around Li^+ for calculations). Regions of the unit cell with minimum energy are regions where Li^+ is likely to travel. Therefore, mapping the minimum energy highlights the likely Li^+ conduction pathways in a given structure. In addition, the information is also useful for estimating the minimum energy required to obtain an infinitely connected surface for Li transport, in at least one direction. The BVEL maps for Li^+ conduction in the six different Bi-O frameworks considered in this study are plotted in Fig. 6. The calculated minimum energy E (also referred to as activation energy) to generate Li^+ conduction paths in at least one direction in these compounds is sufficiently low and ranges from 0.42 eV to 1.65 eV—suggesting Li-Bi-O phases to be good ionic conductors. Among them, Li_7BiO_6 shows the best Li diffusion characteristics with 3D conduction pathways accessible at just 0.42 eV. In fact, these BVEL-based predictions are in agreement with the high ionic conductivities previously reported for Li_7BiO_6 ($5.7 \times 10^{-3} \text{ S/cm}$ at 300 °C) [26]. Other phases showing good Li diffusion capabilities include Li_3BiO_3 with an infinite 2D–3D pathway at 0.45 eV and LiBiO_2 with 2D diffusion

Table 1
Lattice parameters obtained from Le Bail fits of X-ray powder diffraction patterns along with corresponding literature references for the structural models used for each of the Li-Bi-O phases.

Phase	Space group	Lattice parameters	Volume ($\text{\AA}^3$)	Reference
LiBiO_2	<i>Ibam</i> Orthorhombic	$a = 18.00548(10) \text{ \AA}$, $b = 5.18159(3) \text{ \AA}$, $c = 4.97144(3) \text{ \AA}$	463.820(5)	Hoppe et al. [13]
LiBiO_3	<i>Pccn</i> Orthorhombic	$a = 8.82525(6) \text{ \AA}$, $b = 4.91843(3) \text{ \AA}$, $c = 10.69256(6) \text{ \AA}$	464.125(5)	Kumada et al. [14]
Li_3BiO_3	<i>P-1</i> Triclinic	$a = 9.02743(5) \text{ \AA}$, $b = 9.73880(6) \text{ \AA}$, $c = 9.23917(5) \text{ \AA}$, $\alpha = 114.04632(39)^\circ$, $\beta = 89.29499(45)^\circ$, $\gamma = 107.53782(42)^\circ$	701.633(7)	Hoppe et al. [15]
Li_3BiO_4	<i>P4₂/mnm</i> Tetragonal	$a = 8.75298(3) \text{ \AA}$, $c = 4.22469(2) \text{ \AA}$	323.673(2)	Blasse et al. [16]
Li_5BiO_5	<i>Cm</i> Monoclinic	$a = 9.97608(5) \text{ \AA}$, $b = 4.15207(2) \text{ \AA}$, $c = 5.94635(2) \text{ \AA}$, $\beta = 109.48577(33)^\circ$	232.199(2)	Greaves et al. [12]
Li_7BiO_6	<i>R-3</i> Rhombohedral	$a = 5.50567(1) \text{ \AA}$, $c = 15.44446(6) \text{ \AA}$	405.437(2)	Mühle et al. [28]

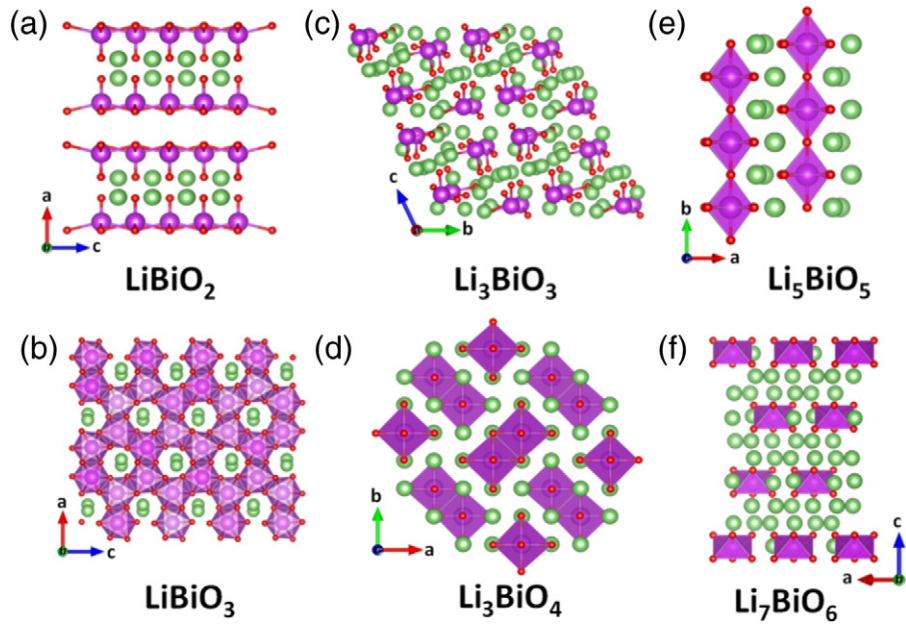


Fig. 2. Crystal structure representations of (a) LiBiO_2 , (b) LiBiO_3 , (c) Li_3BiO_3 , (d) Li_3BiO_4 , (e) Li_5BiO_5 , and (f) Li_7BiO_6 based on previously reported structural models. Li, Bi, and O atoms are represented as green, purple, and red spheres respectively.

paths at 0.65 eV. The remaining Li-Bi-O phases (Li_5BiO_5 , Li_3BiO_4 , and LiBiO_3) require higher activation energies (>1 eV) for Li diffusion and are therefore predicted to have lower Li-ion conductivities.

While the experimental ionic conductivity of some of the Li-Bi-O phases has been previously reported, the values for the other phases are not available. Hence, we measured the ionic conductivities and activation energies of all the phases studied (Figs. 7 and S1), except for LiBiO_3 , which has limited thermal stability (250°C). Here we like to note that our BVLE analysis calculated the highest activation energy (1.66 eV) and thereby predicts the lowest Li-ion conductivity for LiBiO_3 among all of the Li-Bi-O phases studied.

The measured conductivity of LiBiO_2 (2.4×10^{-7} S/cm at 300°C) was lower than the reported 10^{-6} S/cm at 300°C [27]. However, the

observed difference in measured conductivities could simply be a result of differences in sample morphology as anisotropy in ionic conductivity is a possibility in layered LiBiO_2 . In case of Li_5BiO_5 , the measured conductivity of 3.7×10^{-7} S/cm at 300°C was lower than the reported 4.9×10^{-4} S/cm at the same temperature [18]. Further, the original study on Li_5BiO_5 also describes a structural transition around 120°C with activation energies of 0.90 eV and 0.75 eV at lower and higher temperatures respectively. We did not observe the reported transition, which occurs at low temperatures (outside the window of our study, 250°C – 400°C), and consequently obtained a single E_a of 0.24 eV. Lastly in Li_7BiO_6 , our 2.2×10^{-7} S/cm (at 300°C) was lower than the reported 5.7×10^{-3} S/cm at 300°C [26]. The reported high conductivity of Li_7BiO_6 at lower temperatures has been attributed by Mühle et al. [19]

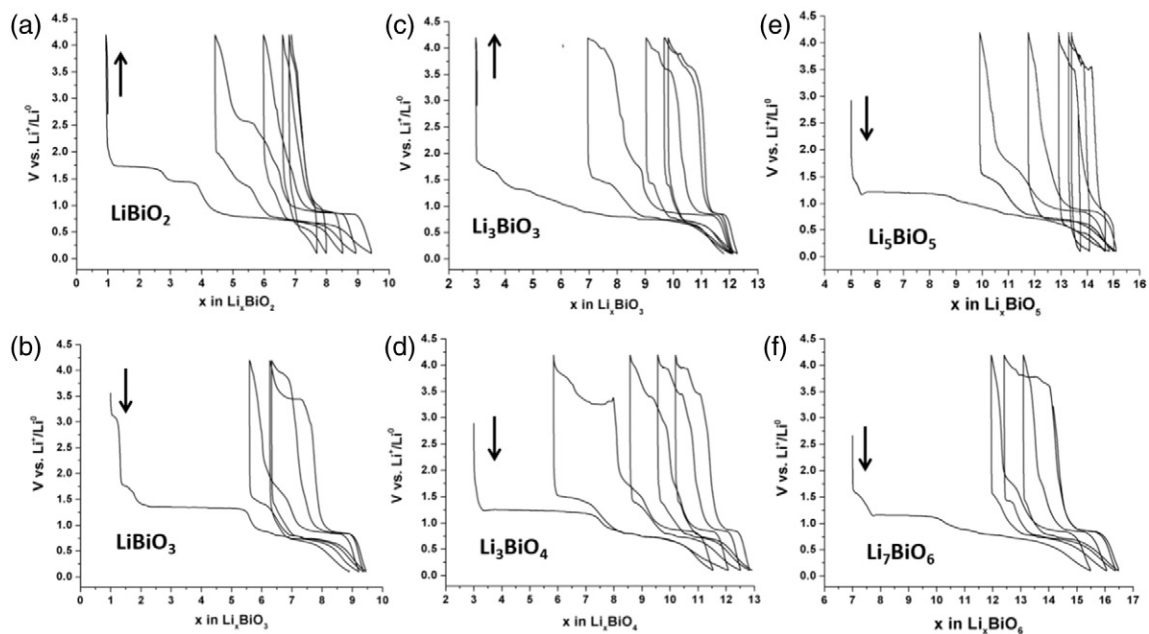


Fig. 3. Cycling of Li-Bi-O phases each with 25 wt% carbon in Swagelok-type half cells vs. Li^+/Li^0 at a rate of C/30; (a) LiBiO_2 , (b) LiBiO_3 , (c) Li_3BiO_3 , (d) Li_3BiO_4 , (e) Li_5BiO_5 , and (f) Li_7BiO_6 . The black arrows indicate beginning of the first cycle.

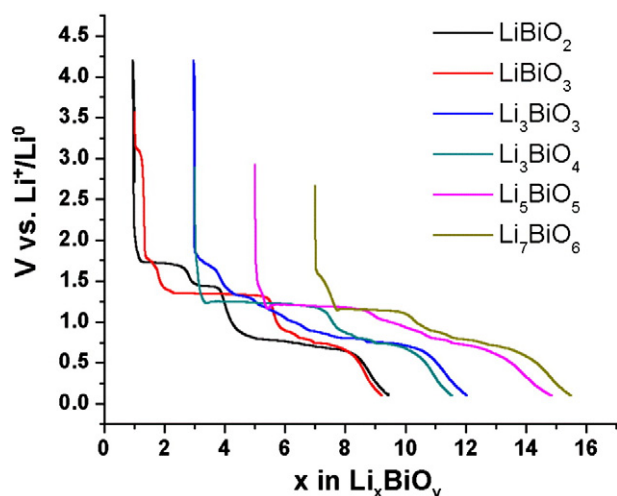


Fig. 4. Conversion reaction observed upon first discharge in all of the Li-Bi-O phases (with 25 wt% carbon) tested in Swagelok-type half cells vs. Li^+/Li^0 at a rate of C/30.

to the triclinic-rhombohedral transition in Li_7BiO_6 . However, our material best fits the rhombohedral model, which could partially explain the lower conductivity value [29]. Further, Li_7BiO_6 has been shown by Nomura et al. [26] to undergo a low temperature transition ($<100^\circ\text{C}$) with $E_a = 0.21$ eV and 0.68 eV at lower and higher temperatures, respectively. This reported transition was below the lower temperature limit of our experimental setup and therefore not observed. However, using data collected between 250°C and 400°C , we obtained a single activation energy of 0.53 eV.

As we are interested in these materials for battery applications, their room temperature conductivity values are most relevant. Hence, we used Arrhenius plots of temperature dependent ionic conductivities to extrapolate for values at 25°C (Table 2). Among the phases studied, LiBiO_2 showed the highest conductivity of 3.8×10^{-8} S/cm at 25°C . The Li-Bi-O phases can be broadly classified into two groups based on

the Bi oxidation state: phases with Bi^{3+} and phases with Bi^{5+} . Unlike the Bi^{5+} oxides (Li_3BiO_4 , Li_5BiO_5 , Li_7BiO_6), the Bi^{3+} (LiBiO_2 and Li_3BiO_3) phases have s^2 lone pair electrons. The presence of a lone pair of electrons would alter the Li local environment and thereby affect the movement of Li^+ in the structure. In fact, such a distinction was observed in the measured activation energies of the Bi^{3+} and Bi^{5+} phases, which were calculated from the slopes of the curves in the Arrhenius plot. The value of activation energy (E_a) indicates the relative ease of Li^+ movement in the structure, but not the specific mechanism involved in the process. Our results show lower average activation energy in Bi^{3+} phases ($0.1\text{--}0.33$ eV) compared to Bi^{5+} phases ($0.37\text{--}0.53$ eV). An exception to this trend was Li_5BiO_5 with E_a of 0.24 eV. The lower E_a in Bi^{3+} phases suggest the lone pair electrons to be advantageous for Li^+ mobility. Here it is worth mentioning that the s^2 lone pair electrons pertaining to Bi^{3+} have previously been used to explain the high oxygen conductivity in $\text{Bi}_4\text{V}_2\text{O}_{11}$ or BIMEVOX-type compounds [29]. Hence, a direct comparison of measured conductivities in oxides with Bi^{3+} and Bi^{5+} may not be suitable.

Looking first at the Bi^{3+} phases, LiBiO_2 and Li_3BiO_3 , have very different crystal structures. LiBiO_2 has a layered structure which allows for 2D mobility of Li^+ ions, while Li_3BiO_3 has a structure that allows for 3D mobility of Li^+ ions. Based on these structural differences, Li_3BiO_3 would be expected to show higher ionic conductivity than LiBiO_2 , and it is also the trend predicted by our BVEL analysis. However our experimental results show lower conductivity for Li_3BiO_3 (2.4×10^{-10} S/cm) compared to LiBiO_2 (3.8×10^{-8} S/cm). On the other hand in the Bi^{5+} phases, based on crystal structure, Li_3BiO_4 with possible 1D mobility of Li^+ would be expected to have lower ionic conductivity compared to Li_7BiO_6 with pseudo-2D Li^+ , and Li_5BiO_5 with 2D Li^+ mobilities. While our predicted trend holds true for Li_5BiO_5 (3.9×10^{-9} S/cm), which has the highest conductivity, the lower conductivity of Li_7BiO_6 (1.1×10^{-11} S/cm) compared to Li_3BiO_4 (4.1×10^{-10} S/cm) is inconsistent with our structure-based predictions.

Generally, our measured conductivity values were lower than those previously reported. However, it is important to recognize that measured values depend on several parameters including phase purity and porosity of the pellet. Conductivity decreases with porosity [30], and our pellets post-sintering (40% porosity) had twice the porosity of pellets used in the previous studies (20% porosity) [26]. Another factor significantly affecting the conductivity is the presence of small quantities of impurity phases. Further, we cannot entirely exclude the possibility of an amorphous phase at the grain boundaries affecting conductivities, especially since we used low temperature synthesis methods for sample preparation. In addition, there is a possibility of generating tiny amounts of secondary phases during conductivity measurements where samples reach temperatures as high as 320°C . We originally tested the conductivity of these materials using gold contacts, which are commonly used as blocking electrodes for ionic conductivity measurements. However, we observed surface color tarnishing of Li_7BiO_6 pellet upon completion of conductivity measurements—indicative of a surface reaction. Hence, we conducted a post-mortem XRD analysis of the measured Li_7BiO_6 pellet, and our results show evolution of new (weak) peaks corresponding to small amounts of LiBiO_2 . While not all pellets investigated in the study showed such obvious signs of chemical reaction, X-ray analysis of pellets of all materials post conductivity measurements confirmed minor surface compositional changes. Although previous studies on ionic conductivity of Li-Bi-O phases used Au-contacts, and the measurements were conducted in temperature ranges similar to ours, surprisingly none of them discuss pellet composition post-measurements. Therefore, it is unclear whether a direct comparison of our measured conductivity values to those previously reported is meaningful. In order to avoid such compositional changes, we used carbon contacts for the ionic conductivity measurements reported herein. Such measurements using carbon contacts represent the overall conductivity (ionic + electronic) of the system. However, we believe the electronic conductivity contribution to be

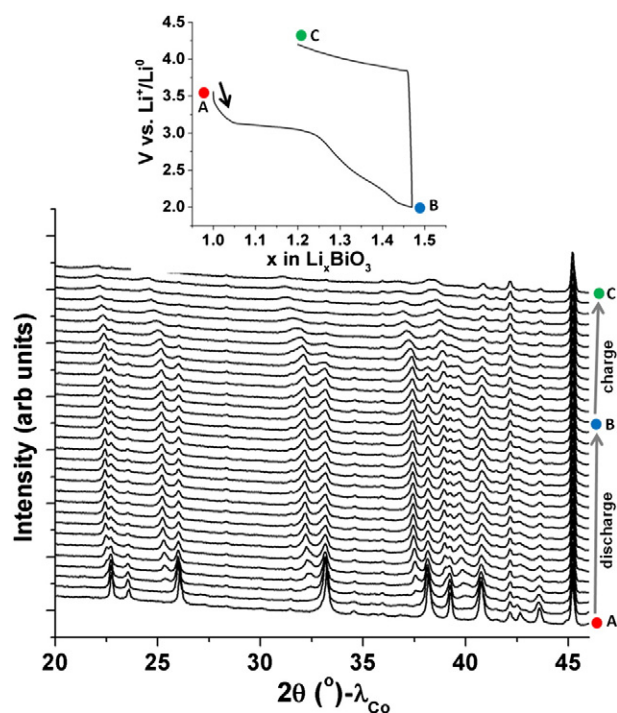


Fig. 5. In situ X-ray diffraction of first discharge-charge of LiBiO_3 (with 25 wt% carbon) vs. Li^+/Li^0 at C/60. The inset shows the corresponding cycling curve.

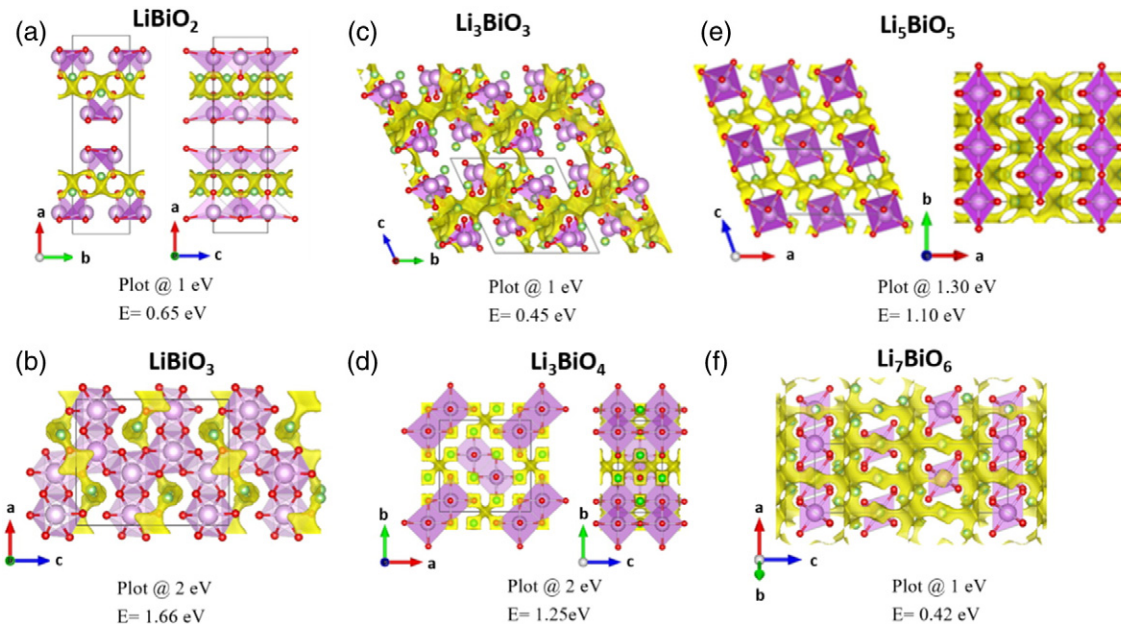


Fig. 6. Bond valence energy landscapes for (a) LiBiO_2 , (b) LiBiO_3 , (c) Li_3BiO_3 , (d) Li_3BiO_4 , (e) Li_5BiO_5 , and (f) Li_7BiO_6 . The minimal energy E required to generate at least one infinitely connected Li-conduction pathway is reported for each compound. The value chosen for plotting the energy iso-surface (colored in yellow) is also indicated (Plot @). Li, Bi, and O atoms are represented as green, purple, and red spheres respectively.

negligible as the very low measured ionic conductivities in these materials are in agreement with their lack of electrochemical activity (i.e. poor Li-ion mobility) when mixed with carbon (to enhance electrical conductivity) and cycled vs. Li^+/Li^0 .

We started our study of Li-Bi-oxides with the hopes of identifying compounds within which $2e^-$ redox occurs due to $\text{Bi}^{3+}/\text{Bi}^{5+}$ transitions. Unfortunately, none of the phases we examined showed any promising electrochemical activity. Nevertheless, the existence of LiBiO_2 – LiBiO_3 and Li_3BiO_3 – Li_3BiO_4 pairs which vary in composition by just one oxygen atom and hence in their Bi oxidation state ($\text{Bi}^{3+}/\text{Bi}^{5+}$) could provide an alternative means to explore the nature of electron transfer going between the Bi^{3+} and Bi^{5+} phases. Identifying mixed-valence intermediate compounds during $\text{LiBiO}_3 \rightarrow \text{LiBiO}_2$ and $\text{Li}_3\text{BiO}_3 \rightarrow \text{Li}_3\text{BiO}_4$ transitions would suggest existence of disproportionation reactions. Hence, we studied these transitions using TGA and *in situ* X-ray diffraction experiments.

TGA analysis of LiBiO_3 under O_2 showed a weight loss of 6 wt% (Fig. S2), which is in good agreement with the expected 6.06 wt% loss for going from LiBiO_3 to LiBiO_2 . LiBiO_2 is thermodynamically more stable than LiBiO_3 ; hence, even upon heating under O_2 , the Bi^{3+} phase is obtained. To get more insight into the $\text{LiBiO}_3 \rightarrow \text{LiBiO}_2$ transition, *in situ* XRD of LiBiO_3 sample heated from 100 °C to 560 °C under N_2 flow was conducted (Fig. 8). Our results clearly show that LiBiO_3 upon heating undergoes decomposition to form Bi_2O_3 and Li_2O (amorphous and not visible by XRD) and the decomposition products then recombine to form LiBiO_2 . No intermediate mixed-valence phase is observed, and when the sample is cooled back down to room temperature, it remains LiBiO_2 . Similarly, heating Li_3BiO_3 under O_2 flow initially resulted in a multi-phase sample. Among the various phases formed, the amount of Li_3BiO_4 phase increased when heated to temperatures greater than 500 °C (Fig. S3), but the experiment did not reveal the emergence of any new intermediate phases with $\text{Bi}^{3+}/\text{Bi}^{5+}$ mixed oxidation state.

Finally, having found no suitable Bi-based two electron systems we expanded the survey to corresponding Li-Sb-O phases. The previously reported Li-Sb-O phases include LiSbO_2 [31], Li_7SbO_6 [19], Li_5SbO_5 [17], Li_3SbO_4 [32], LiSb_3O_8 [33], and LiSbO_3 [34]. In fact, several of these oxides have been studied for Li-ion batteries by Kundu et al. as possible candidates for negative conversion electrodes [35–38]. While some of these phases are isostructural to corresponding Li-Bi-O compositions that we examined (Li_7SbO_6 , Li_5SbO_5 , LiSbO_3), others are not (LiSbO_2 , Li_3SbO_4). However, irrespective of structural correlation, no electrochemical activity was observed when several Li-Sb-O phases prepared by ceramic methods were cycled vs. Li^+/Li^0 in Swagelok-type half cells. A typical cycling curve obtained for these materials is shown in Fig. S4.

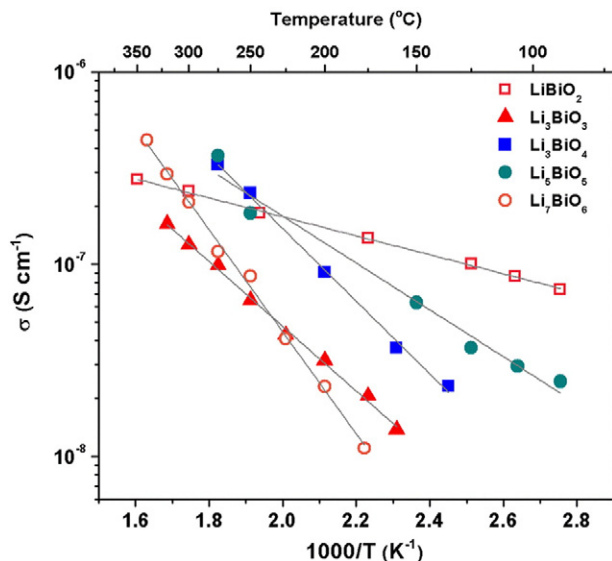


Fig. 7. Temperature dependent ionic conductivity of Li-Bi-O phases.

Table 2

The activation energy and ionic, $c = 15.44446(6)\text{\AA}$ conductivity values for the Li-Bi-oxide phases studied.

Phase	E_a (eV)	σ at 25 °C (S/cm)
Li_7BiO_6	0.53	1.1×10^{-11}
Li_5BiO_5	0.24	3.9×10^{-9}
Li_3BiO_4	0.37	4.1×10^{-10}
Li_3BiO_3	0.33	2.4×10^{-10}
LiBiO_2	0.10	3.8×10^{-8}

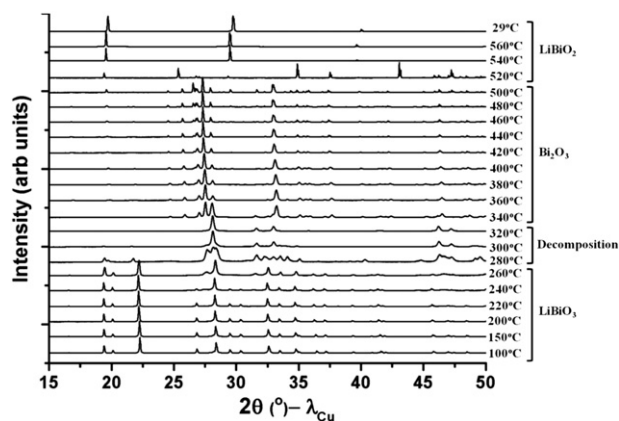


Fig. 8. *In situ* X-ray powder diffraction patterns of LiBiO_3 heated under N_2 from 100 °C to 560 °C and cooled back to 29 °C. LiBiO_3 decomposes to constituent Li_2O and Bi_2O_3 , which then recombine to form LiBiO_2 at higher temperatures.

4. Conclusion

All of the reported Li-Bi-O phases were successfully prepared either by ceramic or by hydrothermal methods and tested for Li-electrochemical activity. None of the phases showed any notable redox activity at higher potentials, while at lower potentials conversion reactions with $\text{Bi}^{3+}/\text{Bi}^{5+} \rightarrow \text{Bi}^0$ was observed. The ionic conductivity of the materials was measured, and among them, LiBiO_2 showed the highest conductivity of 3.8×10^{-8} S/cm at 25 °C. The transformation of $\text{LiBiO}_3 \rightarrow \text{LiBiO}_2$ and $\text{Li}_3\text{BiO}_3 \rightarrow \text{Li}_3\text{BiO}_4$ was observed by *in situ* X-ray diffraction but no intermediate phases with mixed $\text{Bi}^{3+}/\text{Bi}^{5+}$ oxidation states were identified.

Author contributions

CVS and JMT performed the synthesis and electrochemical characterizations, GR conducted the crystal structure and BVOL analyses, RNV assisted with X-ray diffraction studies, and the ionic conductivity measurements were conducted by CVS, CL, and PB.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <http://dx.doi.org/10.1016/j.ssi.2015.10.016>.

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